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AI School: Iteration 1.5

Statistics

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<h4> A little bit of stats </h4>

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Agenda

01. Stats & ML

02. Basics

03. Measures of Central Tendency

04. Measures of Dispersion

05. Outliers

06. Distributions

07. Correlation

08. Stats → ML

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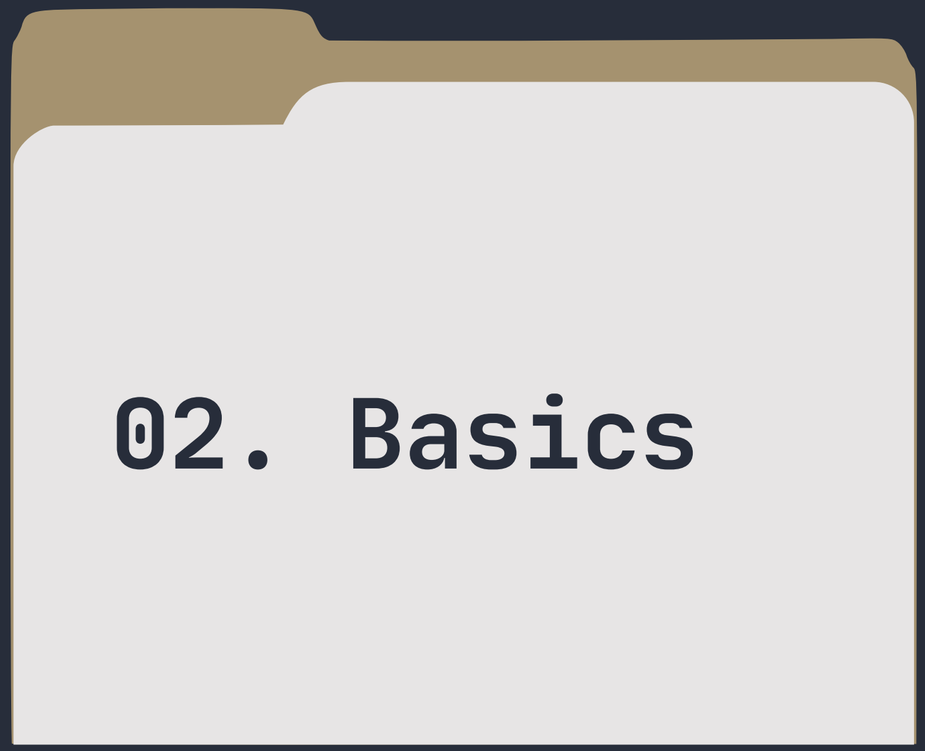
01. Stats & ML

Statistics and machine learning are deeply intertwined in two major ways:

1- Stats is the language of many ML models
Many ML algorithms are built on statistical foundations. Linear regression, Naive Bayes, logistic regression, PCA

2- Stats is how we explore data before training
Before feeding data to any algorithm, we need to understand it. No good model was ever built on poorly understood data.

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02. Basics

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02. Basics: Types of Data

Numerical (Quantitative):
Data that represents measurable quantities.
Can be:

- continuous (height, temperature, salary);
- or discrete (number of items, count of events).

Categorical (Nominal | Qualitative):
Data that represents categories with no natural order.
Examples: gender, country, color, type of animal.

Ordinal:
Data with categories that have a meaningful order
Examples: rating (1-5), shirt size (s < m < l)

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02. Basics: Population vs Sample

Population
The entire group you want to draw conclusions about.
Example: all students in Algeria

Sample
A subset of the population that we actually collect and analyze.
Example: 500 randomly selected students

A good sample is representative: it reflects the population without bias.

👉 **Most real-world datasets are samples, not populations.**

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02. Basics: Descriptive vs Inferential Stats

Descriptive Stats:
Summarizes and describes the data you have.

Answers: *What does my data look like?*

Examples: mean, median, standard deviation, histograms, frequency tables.

Inferential Stats:
Uses a sample to make conclusions about a larger population.

Answers: *What can I conclude beyond my data?*

Examples: hypothesis testing, confidence intervals, p-values.



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03. Measures of Central Tendency

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03. Measures of Central Tendency: Mean

$$\frac{\sum_{i=1}^{n} x_i}{n}$$

The mean is the arithmetic average of all values in a dataset.

Mean = Sum of all values / Number of values

Example: [2, 4, 6, 8, 10] → Mean = 30 / 5 = 6

+ Easy to understand and to compute
_ Highly sensitive to outliers. One extreme value can pull the mean far from the "center" of most data points.

[2, 4, 6, 8, 1000] → Mean = 204 (not representative at all)

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03. Measures of Central Tendency: Median

The median is the middle value when data is sorted in order.

- If the count is odd: take the middle value.
- If the count is even: take the average of the two middle values.

Example: [2, 4, 6, 8, 10] → Median = 6
[2, 4, 6, 8] → Median = (4+6)/2 = 5

\+ Robust to outliers.

👉 Use when outliers are present, data is skewed.
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03. Measures of Central Tendency: Mode

The mode is the most frequently occurring value in a dataset.

A dataset can have:

- No mode – all values appear once
- One mode – unimodal
- Two modes – bimodal
- Multiple modes – multimodal

👉 The only measure that works for categorical data.

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04. Measures of Dispersion

04. Measures of Dispersion: Range

The range is the simplest measure of dispersion.

Range = Maximum value - Minimum value

Example: [3, 7, 2, 9, 1] → Range = 9 - 1 = 8

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04. Measures of Dispersion: Variance

$$: \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}.$$

The average of the squared differences from the mean. Measures how spread out the values are.

- Low variance → values cluster tightly around the mean
- High variance → values are scattered widely

Problem: Variance is in squared units (e.g., meters² for height data), making it hard to interpret directly.

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04. Measures of Dispersion: Standard Deviation

The square root of variance. Brings the measure back to the original unit of the data.

- Low std dev → values cluster tightly around the mean
- High std dev → values are scattered widely

Example: Heights with mean = 170cm, Std Dev = 10cm → most people are within 160–180cm.

👉 Std Dev tells you: *on average, how far is each value from the mean?*

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04. Measures of Dispersion: 5 number summary

The five number summary gives a compact picture of a dataset's spread:

(min, Q1, Q2, Q3, max)

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The five number summary is the foundation of the box plot

#	Statistic	Description
1	**Min**	Smallest value
2	**Q1**	25th percentile — 25% of values fall below
3	**Q2**	50th percentile — the median
4	**Q3**	75th percentile — 75% of values fall below
5	**Max**	Largest value

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04. Measures of Dispersion: IQR

Represents the spread of the middle 50% of the data.

IQR (Interquartile Range) = Q3 - Q1

👉 IQR focuses on the middle 50% — robust to outliers unlike range.

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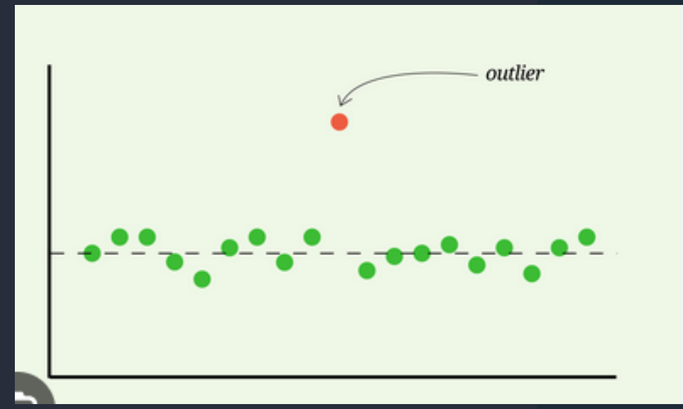
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05. Outliers

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05. Outlier Detection: IQR method

An outlier is a data point that lies abnormally far from the rest of the dataset.



IQR Method:

1. Compute Q1 and Q3
2. Compute IQR = Q3 - Q1
3. Lower bound = Q1 - 1.5 × IQR
4. Upper bound = Q3 + 1.5 × IQR
5. Any value outside [Lower, Upper] is flagged as an outlier

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05. Spurious vs Non-Spurious Outliers

Not all outliers are equal. Before removing them, you need to understand why they exist.

Spurious Outliers (Noise)

Caused by errors: data entry mistakes, sensor malfunctions, corrupted records.
→ These should usually be removed or corrected.

Example: A patient's age recorded as 320. Clearly an error.

Non-Spurious Outliers (Real signal)

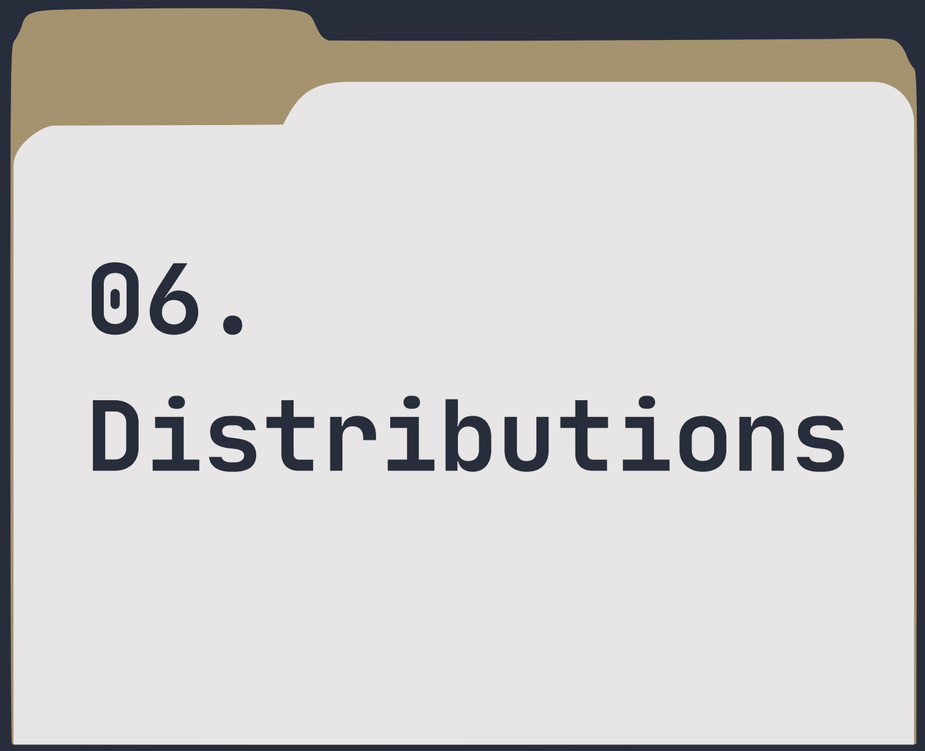
Genuine extreme values that reflect reality.
→ These should be kept and investigated.

Example: A transaction of $1,000,000 in a fraud detection dataset: this could be the fraud you're looking for.

👉 **Blindly removing outliers can destroy the most interesting part of your data.**

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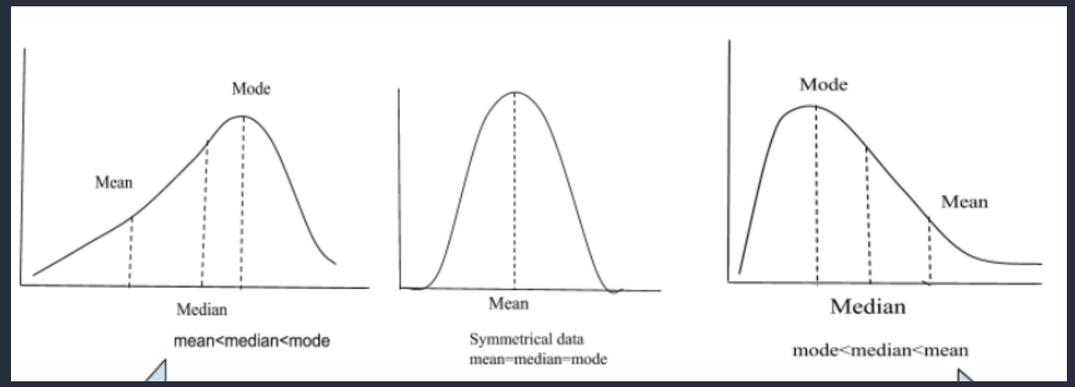
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06. Distributions

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06. Distributions



A distribution shows how values spread across all possible outcomes.

Normal (Gaussian) Distribution

Symmetric, bell-shaped curve. The most important distribution in statistics.

The 68–95–99.7 Rule:

68% of data falls within 1 Std Dev of the mean

95% within 2 Std Dev | 99.7% within 3 Std Dev

Skewness — when the distribution is NOT symmetric

Right-skewed: tail on the right, Mean > Median. Example: income data.

Left-skewed: tail on the left, Mean < Median. Example: exam scores near ceiling.

☞ **Skewness tells you if the mean is being pulled away from the true center.**

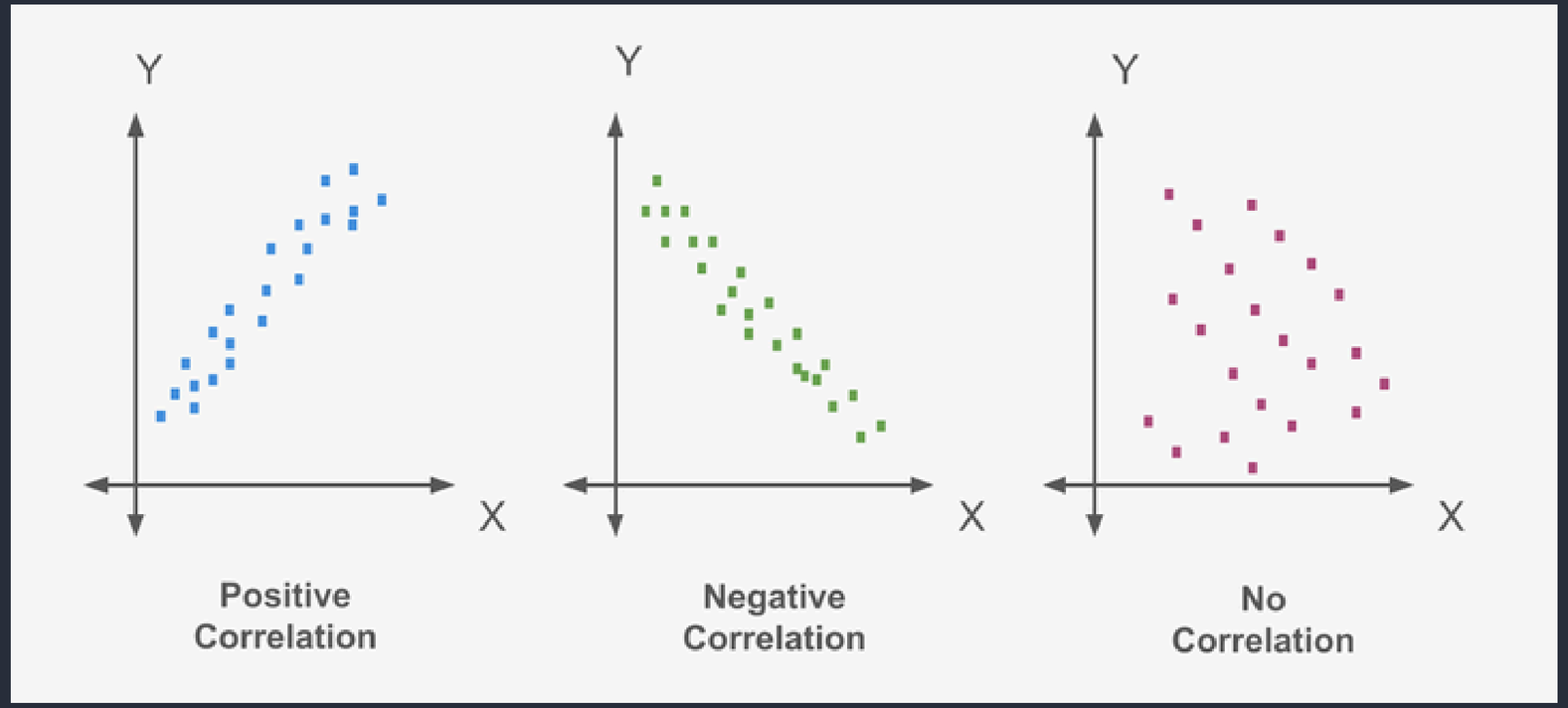
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07. Correlation

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07. Correlation



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07. Correlation: Pearson's R

Measures the strength and direction of the linear relationship between two numerical variables.

Range: $-1 \leq r \leq +1$

r = +1 Perfect positive correlation
r > 0.7 Strong positive | r ≈ 0 No linear correlation
r < −0.7 Strong negative | r = −1 Perfect negative correlation

In pandas (one line!):

```
df.corr()                  # full correlation matrix
df['A'].corr(df['B'])      # between two specific columns
```

☞ **Pearson's r only detects LINEAR relationships.**

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08. Stats → ML

📌 Everything we covered is the backbone of ML

Mean and std dev → Feature scaling
Correlation → Feature selection
Distributions → Model assumptions
Outlier detection → Data cleaning

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Any Questions ?

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Thank you for your attention ✨

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